WORKED SOLUTION

STEP III 2018, Question 11

Mechanics • Slack String Projectile

STEP III 2018, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

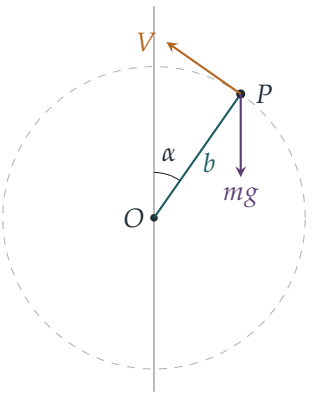
step.maths.org/questions/18-s3-q11

The key idea

The string goes slack at the moment its tension falls to zero, so the radial equation of motion at that instant gives V. After that the particle is an ordinary projectile, and the string becomes taut again when the particle's distance from O returns to b. Working in coordinates centred on O, that condition collapses beautifully, because the projectile starts on the circle moving along the tangent. The final part is then a question of whether the velocity at the jerk points straight along the string.

Solution

We model the particle as a point mass m on a light inextensible string, with no air resistance. Take O as origin, with x horizontal and y vertically upwards, and suppose the particle swings anticlockwise so that it goes slack at the point P on the right of the upward vertical.



At P the string makes angle α with the upward vertical, so $P = (b\sin\alpha,\ b\cos\alpha)$. The

velocity there is perpendicular to the string and directed up and over the top:

$$\mathbf{v}_P = V(-\cos\alpha,\ \sin\alpha).$$

The speed when the string goes slack

Resolve towards O along the string. The tension R acts towards O, and since P is above O the weight has component $mg\cos\alpha$ towards O as well. The particle is moving in a circle of radius b, so

$$R + mg\cos\alpha = \frac{mV^2}{b}.$$

The string becomes slack when $R = 0$, so

$$V^2 = gb\cos\alpha, \qquad V = \sqrt{gb\cos\alpha}.$$

When the string is taut again

Once slack, the particle is a projectile. After time t its position is

$$x = b\sin\alpha - Vt\cos\alpha, \qquad y = b\cos\alpha + Vt\sin\alpha - \tfrac{1}{2}gt^2.$$

The string is taut again when $x^2 + y^2 = b^2$. Expanding,

$$\begin{aligned} x^2 + y^2 &= b^2\sin^2\alpha - 2bVt\sin\alpha\cos\alpha + V^2t^2\cos^2\alpha \\ &\quad + b^2\cos^2\alpha + V^2t^2\sin^2\alpha + \tfrac{1}{4}g^2t^4 + 2bVt\sin\alpha\cos\alpha - bgt^2\cos\alpha - gVt^3\sin\alpha \\ &= b^2 + \left(V^2 - gb\cos\alpha\right)t^2 - gVt^3\sin\alpha + \tfrac{1}{4}g^2t^4. \end{aligned}$$

The terms in t cancel because the particle starts out moving along the tangent, and the terms in t^2 cancel because $V^2 = gb\cos\alpha$. Hence

$$x^2 + y^2 - b^2 = gt^3\left(\tfrac{1}{4}gt - V\sin\alpha\right).$$

For $0 < t < 4V\sin\alpha/g$ this is negative, so the particle is strictly inside the circle and the string stays slack. It is zero again (and becomes positive afterwards) at the time T with

$$gT = 4V\sin\alpha,$$

as required.

The direction of motion just before

At time T the velocity components are

$$\dot{x} = -V\cos\alpha, \qquad \dot{y} = V\sin\alpha - gT = V\sin\alpha - 4V\sin\alpha = -3V\sin\alpha.$$

So the particle is moving down and to the left, and the angle β its path makes with the horizontal satisfies

$$\tan\beta = \frac{|\dot{y}|}{|\dot{x}|} = \frac{3V\sin\alpha}{V\cos\alpha} = 3\tan\alpha,$$

as required.

Coming to rest

Ask yourself What has to be true of the velocity for the particle to stop dead?

The jerk destroys only the component of momentum along the string. The component perpendicular to the string is untouched, so the particle comes to rest if and only if that perpendicular component is already zero, that is, if and only if the velocity just before is parallel to the string $\overrightarrow{OQ}$, where Q is the particle's position at time T. (The velocity is not zero, since $\dot{x} = -V\cos\alpha \neq 0$.)

Using $V^2 = gb\cos\alpha$ and $T = 4V\sin\alpha/g$,

$$x(T) = b\sin\alpha - \frac{4V^2\sin\alpha\cos\alpha}{g} = b\sin\alpha\,(1 - 4\cos^2\alpha),$$
$$y(T) = b\cos\alpha + \frac{4V^2\sin^2\alpha}{g} - \frac{8V^2\sin^2\alpha}{g} = b\cos\alpha\,(1 - 4\sin^2\alpha).$$

The vectors (x, y) and $(\dot{x}, \dot{y})$ are parallel if and only if $x\dot{y} - y\dot{x} = 0$. Dividing by $-bV$,

$$3\sin^2\alpha\,(1 - 4\cos^2\alpha) - \cos^2\alpha\,(1 - 4\sin^2\alpha) = 0.$$

Write $S = \sin^2\alpha$, so $\cos^2\alpha = 1 - S$ and $1 - 4\cos^2\alpha = 4S - 3$:

$$3S(4S - 3) - (1 - S)(1 - 4S) = 0$$
$$12S^2 - 9S - (1 - 5S + 4S^2) = 0$$
$$8S^2 - 4S - 1 = 0,$$

so $S = \frac{1}{16}(4 \pm \sqrt{16 + 32}) = \frac{1}{4}(1 \pm \sqrt{3})$. Since $S = \sin^2\alpha > 0$, the negative root is impossible, and the remaining root lies in $(0, 1)$, so it corresponds to a genuine acute angle. Every step above is reversible, so the particle comes instantaneously to rest when the string becomes taut if and only if

$$\sin^2\alpha = \frac{1 + \sqrt{3}}{4}.$$

Checks

- The formula $V^2 = gb\cos\alpha$ gives $V \to 0$ as $\alpha \to \frac{1}{2}\pi$: the string goes slack at the level of O only if the particle has stopped there, which is right.
- With $\sin^2\alpha = \frac{1}{4}(1 + \sqrt{3})$ we get $\alpha \approx 55.7°$ and $\cos^2\alpha = \frac{1}{4}(3 - \sqrt{3})$, so $Q = \big(b\sin\alpha\,(\sqrt{3} - 2),\ -\sqrt{3}\,b\cos\alpha\big)$, below and to the left of O. The slope of OQ is $\sqrt{3}\cot\alpha/(2 - \sqrt{3})$ and the slope of the path is $3\tan\alpha$. These agree, because

$$3(2 - \sqrt{3})\tan^2\alpha = \frac{3(2 - \sqrt{3})(1 + \sqrt{3})}{3 - \sqrt{3}} = \frac{3(\sqrt{3} - 1)}{\sqrt{3}(\sqrt{3} - 1)} = \sqrt{3}.$$

  A computer simulation of the motion confirmed all three results.

Common slips

- Using energy alone to find V: energy links V to the initial speed, but it is the zero-tension condition that fixes V in terms of b, g and α.
- Getting the velocity direction at P wrong. It is perpendicular to OP, so it makes angle α with the horizontal, not with the vertical.
- Forgetting to justify that T is the *first* time the string is taut; the factorisation $gt^3(\frac{1}{4}gt - V\sin\alpha)$ shows it is.

Beyond STEP

A string can pull but not push, so a slack-string problem is really a constrained system whose constraint switches on and off; the same idea of one-sided constraints appears in contact mechanics and in Lagrangian mechanics with inequality constraints. The sudden jerk is an impulsive constraint force, which is treated in first-year courses alongside collisions and the coefficient of restitution. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops both circular motion and impulses carefully.

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